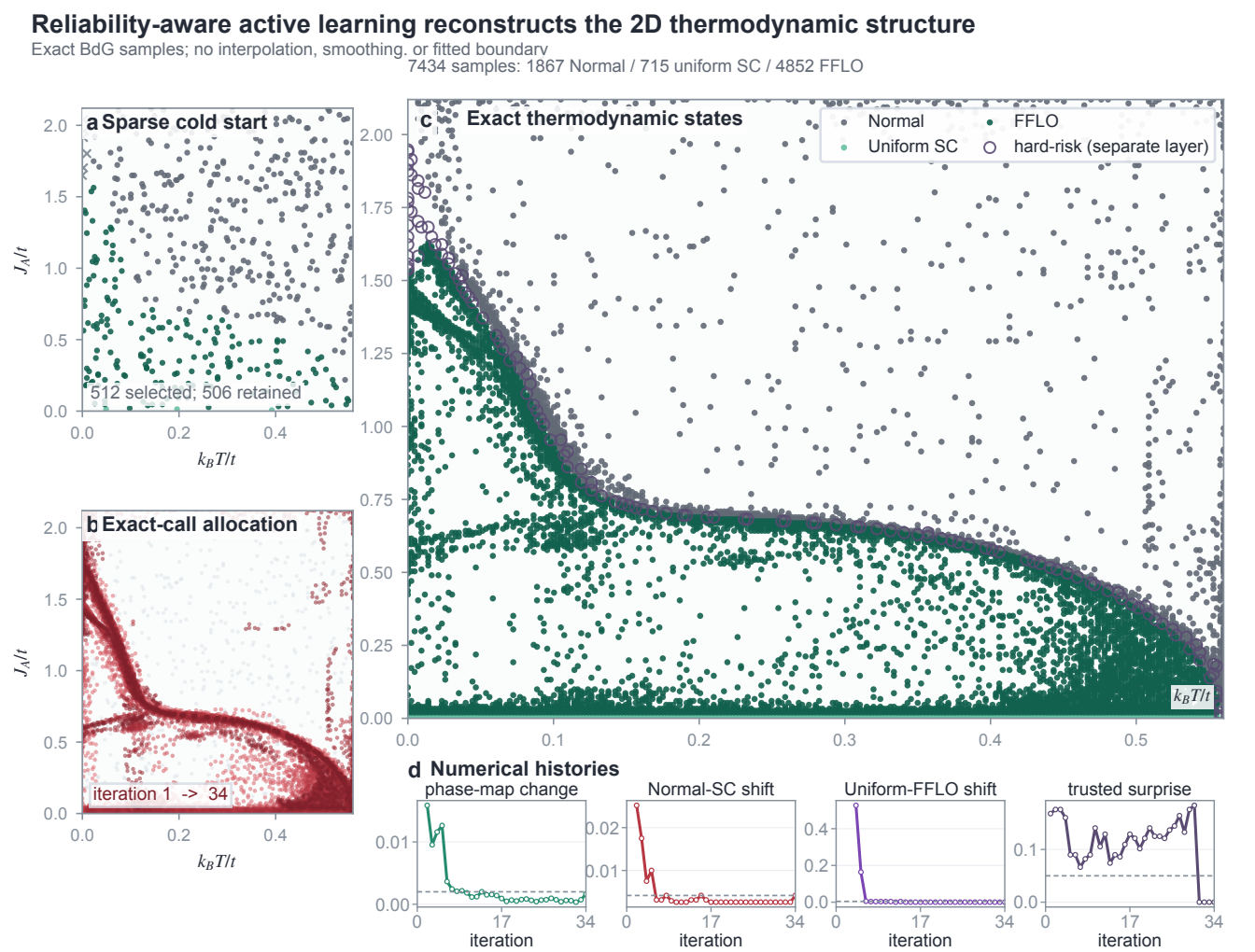

Reliability-aware active learning reconstructs the 2D thermodynamic structure
Exact BdG samples; no interpolation, smoothing, or fitted boundary
7434 samples: 1867 Normal / 715 uniform SC / 4852 FFLO
a Sparse cold start
512 selected; 506 retained
b Exact-call allocation
iteration 1 -> 34
c Exact thermodynamic states
Normal
Uniform SC
FFLO
hard-risk (separate layer)
J_A/t
k_BT/t
d Numerical histories
phase-map change
Normal-SC shift
Uniform-FFLO shift
trusted surprise
iteration